

Bhargav Nallan Chakravartula

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EDUCATION

Arizona State University (ASU)

M.S. Electrical Engineering (GPA: 4.0)

Tempe, AZ

Jan. 2024 – Aug. 2026

IIIT D&M,Kancheepuram

B&M Technology Electronics & Communications Engineering (GPA: 8.0)

Chennai, India

Jul. 2016 – May 2021

TECHNICAL SKILLS

Languages : Verilog, C/C++, Python, SystemVerilog, TCL, VHDL, Assembly

Tools : Cadence Design Systems (Virtuoso, Innovus, OrCAD), Xilinx Vivado, Mentor Questa Sim, Solidworks, Synopsys (HSpice, Design Compiler (DC), IC Validator, StarRC, IC Compiler, IC Validator, PrimeTime), Mentor Graphics (Calibre, Modelsim), Open source (OpenRoad, Ngspice), LTspice, Eagle, Proteus, Arduino IDE, MATLAB

Hardware : Field-Programmable Gate Arrays (FPGA), Printed Circuit Board (PCB), Arduino UNO

Laboratory Techniques: Signal Generator, Digital Storage Oscilloscope, Digital Multimeter, Spectrum Analyzers

Other: Operating Systems (Linux/UNIX, Windows), Static Timing Analysis, Digital Circuit Analysis

EXPERIENCE

Foundry Technology Summer Intern

Skyworks Solutions, Inc

May 2025 – Aug. 2025

Austin, TX

- Achieved 25% reduction in PO-related delays and \$2M in annual cost avoidance by eliminating redundant mask generation (\$10K per mask per project) through workflow automation and early data validation.
- Developed VBA and Python automation to generate photolithography mask lists (manufacturing stencils) by analyzing design and finalized layouts, enabling early cost estimation and accurate post-layout validation.
- Integrated GlobalFoundries tool into the Skyworks's foundry workflow to automatically extract device and mask information from layout, reducing manual processing, improving submission accuracy and turn around time.
- Developed Python+KLayout automation to compare foundry(MEBES) data (e.g., Tower Semiconductor) with mask layouts, generating visual diffs & error reports; built a multi-portal tracking tool for delivering weekly reports (25 TO) enabling cross-functional coordination (CAD, Foundry Ops, Design & Field Apps) ensuring on-time multi-fab transfers.

Sr Engineer Tapeout Engineering

GLOBALFOUNDRIES Engineering Pvt. Ltd

Jul. 2021 – Dec. 2023

Bangalore, India

- Executed 300+ tapeouts (CAN transceivers, 5G, RF, & PWM chips) for the Tapeout Logistics and Integration (TLI) team despite customer Design Rule Checks (DRC) constraints - ensuring on-time delivery & savings.
- Reduced the cost per tapeout (TO) by 2% (\$700K to \$686K) by erasing an alignment mask from the stack.
- Optimized the Tapeout 4.0 (for CSOI8SW) intake process by enforcing die information to prevent holds and accelerate turnaround time; validated 50+ TO; deployed Mock TO & DRC to reduce resubmits, securing pSemi & Archiwave TOs.
- Earned an Appreciation Award by leading the TLI India night team for Qorvo & Broadcom; Inherited spec releases to Agile PLM tool from the US team like: Expected Change Orders & Component / Device Compliance Specs.
- Delivered tailored DRC solutions (mask rework, split TOs, waivers) to 20+ clients, including modifying Siemens Calibre automation to suppress non-critical density violations across designs.

ACADEMIC EXPERIENCE

Teaching Assistant

Arizona State University

Jan. 2025 – May. 2026

Tempe, AZ

- Led graduate VLSI and Digital Systems labs, developing Integrate & Fire Neuron, 4-bit Adder and Graph Convolutional Network designs adopted as coursework foundations; taught semiconductor device simulation in Silvaco (MOSFET, MESFET, HEMT) for characteristic extraction and band-structure, & drift-diffusion modeling in MATLAB.

Research Assistant

Prof. Dragica Vasileska, ASU

Dec. 2024 – May. 2026

Tempe, AZ

- Directed senior undergraduates in developing a Schrödinger equation solver using shooting and matching methods to compute discrete energy eigenvalues and eigenfunctions for finite square wells; developed bulk Monte Carlo silicon simulators including an ensemble bulk phonon and a two-particle electron-electron scattering + phonon scattering model.

PROJECTS

- Implemented E-K relationship for Si, Ge, GaAs, and AlxGa1-xAs using the empirical pseudopotential method. | [Link](#)
- Modeled a memristor-based temperature sensor in MATLAB; using write time variations between 300K-600K. | [Link](#)
- Designed a Graph Convolutional Network in Verilog & performed APR using Synopsys DC & Cadence Innovus. | [Link](#)
- Developed a 1D model to analyze equilibrium conditions and IV characteristics using drift-diffusion equations. | [Link](#)
- Created a layout for a 1.5GHz Integrate & Fire Neuron using a carry-save adder for four 4-bit weighted inputs. | [Link](#)